Reset-free syndrome extraction on IBM heavy-hex

Corrected report · 15 October 2026 · replaces the 14 October version (the original is not published)

Headline. Removing mid-circuit resets from syndrome extraction for IBM's heavy-hex subsystem code (d=3) lowered the Z-basis logical error per round by ×1.66–2.07 on ibm_kingston in four runs, every one 2σ-separated, including a clean pre-registered rerun on 15 October (×1.72, 10,000 shots). The X-basis gain is smaller (×1.13–1.42). On ibm_marrakesh the direction held but was not significant. Break-even is not reached: the logical qubit lives 0.092× (Z) and 0.136× (X) as long as the best usable physical qubit in the clean test (lifetimes defined in section 3). **What changed from the 14 October report:** six pre-registered predictions that failed are now shown as failures. The 14 October break-even pass was obtained post hoc and is now labelled that way; the 15 October rerun tested it cleanly. The scaling (Λ) claim is restricted to simulation.

1. Corrections to the 14 October report

#	14 Oct report said	What the notebook (Qmem2) shows	Correction
1	Break-even 0.15× (Z) / 0.18× (X), "Measured"	The pre-registered check printed **P16 FAILS**: the probe ranked dead qubit q146 (≈50% readout, fitted T1 2696 µs) as the best physical qubit. Z passed only after q146 was excluded *after* the result.	Labelled post hoc. Retested cleanly on 15 Oct (section 3).
2	"Failures are listed with the same weight as successes"	P3, P4, P6, P10, P13, P15 and P16 printed FAILS. P4 was reported as failed; P10, P13 and P15 were shown as "Mixed"; P3 and P6 were not mentioned; P16 was passed post hoc.	Full verdict table in section 2.
3	"Bigger code helps (Λ>1): never, sim or hardware"	No d=5 heavy-hex job was ever run on hardware.	Λ<1 is a simulation result only.
4	Best error 3.3% per round (no source)	Probe-placement run 1 (job db7rid8r…): 3.33% [2.74, 3.92]. Best tabulated compression arm: 3.67% [3.34, 4.00].	Source added; the wide interval is stated.
5	Shot counts for the compression runs not stated	5,000 shots per circuit (H11, R16, H18, H21, H25). The DD job used 10,000.	Stated.

6	Stale calibration "16–24 of 156 qubits"	Two probes printed 24 and 16; the third printed no count. The 15 Oct probe (Qmem3 cell R5) found 8.	Counts per probe.
7	DD table "error per round", 14–20%	Computed as $1-(1-p)^{1/3}$ with p capped at 0.4999; 3-round errors reached ≈50% (saturated). This is a different definition from the ε fit $F=A(1-2\varepsilon)^r$ used elsewhere, and the cap truncates one interval: 20.53% [20.01, 20.62].	Raw data: without DD the 3-round logical error was 45–50% (X Qiskit 49.8%, near a coin flip), with DD 37–43%. DD lowers the error in 4 of 4 arms (significant at 10,000 shots), but the arms were saturated, so the ×1.21–1.27 size is not meaningful and is withdrawn.
8	"Model predicted the X gain closely"	Holds for run 1 only (×1.45–1.48 predicted vs ×1.42). Runs 2–3: ×1.26, ×1.21.	Scoped to run 1.

2. Every pre-registered prediction, 14–15 October

#	Prediction	Result	Verdict
P1	Tournament placement beats Qiskit (Marrakesh, DD off/on, both bases)	4/4 lower (direction only: same saturated per-round metric as item 7)	HOLDS
P2	X–X DD lowers error (Marrakesh)	4/4 lower; magnitude not meaningful (saturated, item 7)	HOLDS
P3	Best Z ≤ 6% per round (DD job)	14.22% (saturated metric, see item 7)	FAILS
P4	Staggered DD beats uniform DD	2 of 4	FAILS
P5	Tournament beats Qiskit in ≥3 of 4 (Kingston)	3 of 4	HOLDS
P6	Best Z ε ≤ 6% per round	7.48%	FAILS
P7, P8	Compression beats baseline in both bases; Z gain ≥ ×1.3, X ≥ ×1.2 (Kingston run 1)	Z ×1.70, X ×1.42, separated	HOLDS
P9	Kingston replication, 2σ separated	Z ×1.66, X ×1.26	HOLDS
P10	New placement beats earlier, both bases	Z ×0.92, X ×1.18, overlapping	FAILS
P11	Probe-placed beats calibration-placed	Z ×1.90, X ×1.11	HOLDS
P12	Compressed beats baseline (run 3)	Z ×2.07, X ×1.21	HOLDS
P13	Probe placement beats compressed by ×1.3+ in Z	Z ×0.87	FAILS
P14	Best Z ε ≤ 4.5%	3.67%	HOLDS

P15	Live judge ≥ calibration placement, both bases	Z ×1.22, X ×0.79	**FAILS**
P16	Break-even ratio 0.15–0.6×, both bases	Z 0.00× (dead q146 chosen), X 0.18×	**FAILS**
15 October clean rerun (job db88kdgr1c5c738lveeg) — pre-registered 08:09:25 UTC, sha256 6d92b8ad…			
P17	Compression beats baseline, both bases, 2σ separated, 10,000 shots	Z ×1.72, X ×1.13	**HOLDS**
P18	Break-even not reached (upper 2σ bound < 1)	Z 0.096× (≤0.106) as computed; 0.092× with the exact DD model. X 0.136× (≤0.149)	**HOLDS**
P19	Ratio inside 0.10–0.35 in both bases	Z 0.096× (0.092× exact model)	**FAILS**
P20	No reference qubit errs >10% at its shortest wait	1.2–6.0%	**HOLDS**
P21	Live-probe model within ±30% (4 arms)	all 4 within. Caveat: the probe prediction was decoded with its own model, while the hardware data was decoded with the calibration model (found in code review; fixed for the daily study).	**HOLDS**

Definitions and verdicts are quoted from the printed output of Qmem2 and Qmem3.

3. Clean rerun, 15 October 2026 (Qmem3)

Design, fixed before submission: live probe of all 156 qubits (job db88jo6r6oqs7388vrs0). A qubit is unusable if its mean readout error is above 10%, either one-sided readout error is above 15%, or its T1 is not identifiable or above 2 ms. The filter excluded **q113, q120, q121 and q146**, and q146 was still dead (50.0% readout). Zero-SWAP placement used calibration scoring and never touched an excluded qubit. One logical arm per basis (compressed) was fixed in advance for break-even. Physical references were the 3 usable qubits with the longest probe T1 (Z) or T2 (X), idled with X–X DD for 7–400 µs. 30 circuits × 10,000 shots; 99 s of QPU time.

Arm	ε per round (2σ)	Model, calibration	Model, live probe
Z baseline	8.03% [7.71, 8.36]	5.13%	8.70%
Z reset-free	4.68% [4.43, 4.93]	2.21%	3.97%
X baseline	6.39% [6.06, 6.72]	4.17%	7.01%
X reset-free	5.66% [5.36, 5.96]	3.57%	5.18%

Basis	Logical lifetime (µs)	Best physical (µs)	Ratio (upper bound)
Z	50 [48, 53]	q155: 524 [505, 544] symmetric fit; T1 543 [522, 565] exact DD model	0.096× (0.106×); 0.092× exact model
X	60 [57, 63]	q32 / q149: 440	0.136× (0.149×)

The calibration-fed model was again optimistic, by a factor of 1.5–2.1. The probe-fed model stayed within ±30% in all four arms. Physical lifetimes now have ±4% bands, because the waits reach 400 µs; on 14 October they stopped at ≈29 µs.

Lifetime definitions and a re-check

The logical lifetime τ_L is defined by $1 - 2p_L(t) = A \cdot \exp(-t/\tau_L)$. Physical references idled under the same X–X DD (pulses at ¼ and ¾ of each wait). Z references start in $|1\rangle$: the qubit sits in $|1\rangle$ only half the time, so $1 - 2P(0)$ decays close to exponentially, and the symmetric fit recovers T1 within about 5%. A code review raised the concern that a $|1\rangle$ reference decays towards $P(0) = 1$, which would make the symmetric fit report about 0.4 T1. That is true without DD, but not for this sequence. Fitting the exact pulse-sequence model to the raw data (`recompute_breakeven.py`) gives T1 = 543 µs [522, 565] for q155. So the corrected Z ratio is **0.092×**, against 0.096× from the pre-registered code. P18 holds and P19 fails under both.

4. Compression gain across all hardware runs

Run	Chip	Shots	Z gain	X gain	2σ separated
Kingston 1 (db7r4per…)	ibm_kingston	5,000	×1.70	×1.42	yes
Kingston 2 (db7rf1gr…)	ibm_kingston	5,000	×1.66	×1.26	yes
Kingston 3 (db7s2dar…)*	ibm_kingston	5,000	×2.07	×1.21	yes
Marrakesh (db7rbner…)	ibm_marrakesh	5,000	×1.07	×1.05	no

Kingston 4 (db88kdgr…), pre-registered	ibm_kingston	10,000	×1.72	×1.13	yes

*Run 3 also changed the placement (probe-based), so it is not a pure replication.

5. What stands, what does not

- **Stands:** dropping resets (flags kept in X) lowers Z-basis logical error by about ×1.7 on Kingston, replicated four times. X–X DD lowers the error (4/4 on Marrakesh, direction only; arms were near saturation). The calibration-fed model is optimistic and a seconds-long live probe fixes most of that. Break-even is far away: 0.092× (Z) and 0.136× (X) in the clean test.
- **Does not stand:** the 14 October break-even "pass" (post hoc); placement and judging methods predicting their own direction (P10, P13, P15 fail); "Λ<1 on hardware" (never run at d=5); DD per-round values compared with compression ε.
- **Still open:** d=5 on hardware (QPU budget); multi-day replication (hhmem2 automates it); a stronger decoder; Ramsey vs echo to explain the DD gain.

6. Verified against IBM raw data

All 9 hardware jobs were re-fetched from IBM Quantum on 15 Oct and re-analysed with independent bit-parsing and fitting code (`04_hardware_runs/INDEPENDENT_REDERIVATION.json`). IBM's job records confirm the backend (ibm_kingston or ibm_marrakesh), the creation time (14 Oct 19:14–20:57 EEST; 15 Oct 11:09 EEST), status DONE, and the shot count (5,000 per circuit, or 10,000 for the DD job and the 15 Oct rerun). Every ε in sections 1–4 was reproduced to two decimals, including all 8 arms of the DD-type job db7qqker (P6: 7.48%, Z tournament / staggered DD). An earlier version of the re-derivation merged the uniform-DD and staggered-DD arms of that job; this was fixed. QPU time per job (IBM-reported): 19–47 s on 14 Oct; 99 s for the rerun.

7. Reproducibility

Notebooks: Qmem2 (14 Oct record, untouched) and Qmem3 (15 Oct rerun). Code: `hhmem.py` and `hhmem2.py`, which adds the probe, dead-qubit filter, pre-registration and verdicts. Dry run (`dry_run.py`) on FakeKingston with an injected dead q146: the filter excluded it. Control test with the filter disabled: q146 was chosen again and P20 printed FAILS. Raw plan, probe and verdict files are in `04_hardware_runs`. All data can be retrieved from IBM Quantum by job ID.